

Name:

Sole Visions

Date: 6/3/20

Year 9 Mathematics Term 1 2020
Test 1: Algebra and Indices

Time allowed: 50 minutes

No calculators allowed

Resources Allowed:

- 1 piece of A4 paper notes
- Pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

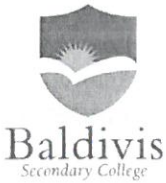
- **Show all your working clearly.** Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks.

- It is recommended that you **do not use pencil**, except in diagrams.

Total Marks:

/50

%



Question 1

[6 marks: 1, 1, 1, 1, 1, 1]

Simplify:

a) $7y + 9y$
 $= 16y$ ✓

d) $7b - 8a + 5b + 10a$
 $= 2a + 12b$ ✓

b) $15p - 5p$
 $= 10p$ ✓

e) $8m \times 7n$
 $= 56mn$ ✓

c) $-3a + 2a - 5a$
 $= -6a$ ✓

f) $9c \times 6c$
 $= 54c^2$ ✓

6

6

Question 2

[6 marks: 1, 2, 1, 1, 1]

Evaluate the following:

a) $7^2 = 49$ ✓

d) $\frac{5^7}{5^2}$ (Leave your answer in expanded form)

$= 5^5$ ✓

b) $(-3)^3 = -27$ ✓

e) $(7^4)^3$ (Leave your answer in index form)

$= 7^{12}$ ✓

c) $2^4 \times 2^3$ (Leave your answer in index form)

$= 2^7$ ✓

6

Question 3

[10 marks: 1, 2, 1, 2, 2, 2]

Show use of the index laws or expand and fully simplify each of the following.

Express your answer in **index form with positive indices**:

a) $x^3 \times x^5 = x^8$ ✓

b) $(2x^3)^2 = 4x^6$ ✓ ✓ or $2^2 x^6$

c) $5x^0 = 5$ ✓

d) $\frac{10x^{15}}{4x^4} = \frac{5x^{11}}{2}$ ✓ ✓ or $2.5x^{11}$

e) $2bc^2 \times 5b^2c^3 = 10b^3c^5$ ✓ ✓

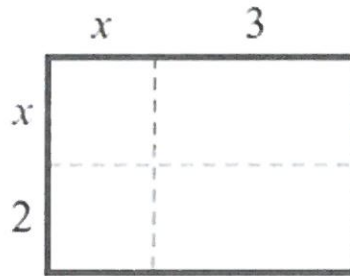
10

f) $\frac{16p^4q^5}{24pq^3} = \frac{2p^3q^2}{3}$ ✓ ✓

16

Question 4**[3 marks]**

The diagram below shows the dimensions of a rectangle in terms of the variable x .



This area of this rectangle can be:

$$\begin{aligned}
 (x+2)(x+3) &= \underline{x^2} + 3x + \underline{2x} + \underline{6} \quad \checkmark \\
 &= \underline{x^2} + \underline{5x} + \underline{6} \quad \checkmark \checkmark
 \end{aligned}$$

3

Question 5**[9 marks: 1, 2, 2, 2, 2]**

Expand and simplify the following by collecting like terms and simplifying where possible.

a) $2(3y - 4) = 6y - 8 \quad \checkmark$

d) $(x+8)(x+7) = x^2 + 7x + 8x + 56 \quad \checkmark$
 $= x^2 + 15x + 56 \quad \checkmark$

b) $5(m-3) + 3m$
 $= 5m - 15 + 3m \quad \checkmark$
 $= 8m - 15 \quad \checkmark$

e) $(n-12)(n+1)$
 $= n^2 + n - 12n - 12 \quad \checkmark$
 $= n^2 - 11n - 12 \quad \checkmark$

c) $2(u-3) - 4(u+5f)$
 $= 2u - 6 - 4u - 20f \quad \checkmark$
 $= -2u - 20f - 6 \quad \checkmark$

9

(12)

Question 6

[13 marks: 2, 2, 2, 3, 4]

Factorise the following expressions and simplify fully:

a) $4b - ab$

$$= b(4 - a) \checkmark \checkmark$$

d) $6a + 2 + 3ab + b$

$$= 2(3a + 1) + b(3a + 1) \checkmark$$

$$= (3a + 1)(2 + b) \checkmark$$

b) $12y^2 + 6y$

$$= 2y(6y + 3) \checkmark \checkmark$$

e) $\frac{5a^2 - 5}{15a^2 - 15}$

$$= \frac{\text{factorise the numerator}}{\text{factorise the denominator}} = \frac{5(a^2 - 1)}{15(a^2 - 1)} \checkmark$$

$$= \text{simplify by cancelling} = \frac{1}{3} \checkmark$$

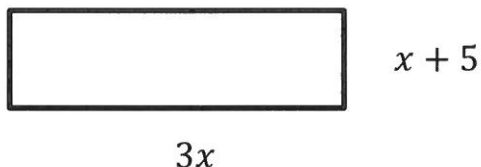
c) $m(y + 2) + (y + 2)$

$$= (y + 2)(m + 1) \checkmark \checkmark$$

Question 7

[3 marks: 1, 2]

A rectangular field is $3x$ m long and $(x + 5)$ m wide.



a) Find the expanded expression for the **area** of the field in terms of x .

$$A = 3x(x + 5) \checkmark$$

b) The length of the fence surrounding the field can be calculated using the **perimeter** of the rectangular field. Give an expression for the perimeter of the field in terms of x , simplifying terms where possible.

$$p = 3x + 3x + x + 5 + x + 5 \checkmark$$

$$= 8x + 10 \checkmark$$

END OF TEST

16